

Database Backup Configuration Guide

System Version: v8.1 (Base Sepolia & Hoodi Network)

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This document details the automated backup architecture and manual recovery procedures for the `sx_prediction_db` PostgreSQL database running in the SX Prediction Marketplace environment.

1. Backup Architecture

The backup strategy utilizes PostgreSQL's native `pg_dump` utility to execute hot backups without taking the database offline. The process is managed by an automated shell script that interfaces with the running Docker container and handles file rotation.

Key Policy:

- **Frequency:** Scheduled daily via Cron.
- **Format:** PostgreSQL Plain SQL text dump (facilitates easy inspection and partial restore).
- **Retention:** Automatically purges backups older than 7 days to preserve disk space.

2. Backup Script (`backup.sh`)

The core backup logic is defined in the script at `backend/scripts/backup.sh`.

Here is the configuration:

```
#!/bin/bash

# Backup script for SX Prediction DB


BACKUP_DIR="/var/backups/postgres"

mkdir -p "$BACKUP_DIR"


TIMESTAMP=$(date +%Y-%m-%d_%H-%M-%S)

BACKUP_FILE="$BACKUP_DIR/sx_prediction_db_${TIMESTAMP}.sql"


echo "Starting database backup at $TIMESTAMP..."


# Execute pg_dump inside the docker container


docker exec sx-postgres pg_dump -U postgres sx_prediction_db >
"$BACKUP_FILE"
```

```
if [ $? -eq 0 ]; then

    echo "Backup successfully created: $BACKUP_FILE"

else

    echo "Backup failed!" >&2

    exit 1

fi


# Clean up backups older than 7 days

find "$BACKUP_DIR" -type f -name "*.sql" -mtime +7 -delete


echo "Old backups cleaned up."
```

3. Automation (Cron Job Integration)

To automate this process, schedule the script via the system's cron daemon on the host. Run `crontab -e` and add the following entry to execute the backup every day at 2:00 AM:

```
0 2 * * * /bin/bash
/home/sanjeev/2026/project-9/backend/scripts/backup.sh >>
/var/log/db_backup.log 2>&1
```

4. Restore Procedure

To restore a database snapshot, use the `psql` utility to pipe the SQL file back into the database instance. Ensure the database target is clean before running the restore.

```
# Drop and recreate the target database

docker exec -it sx-postgres dropdb -U postgres sx_prediction_db

docker exec -it sx-postgres createdb -U postgres sx_prediction_db


# Restore the dump file

docker exec -i sx-postgres psql -U postgres sx_prediction_db <
/var/backups/postgres/sx_prediction_db_YYYY-MM-DD_HH-MM-SS.sql
```